

MAFIKENG, South Africa

WMO: 682420

Lat: 25.8036S

Lon: 25.5428E

Elev: 1281

StdP: 86.85

Time Zone: 2.00 (E02)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.1	2.8	-15.8	1.1	19.3	-13.0	1.4	16.6	10.9	16.3	9.6	15.2	2.7	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.6	34.9	16.2	33.6	16.3	32.2	16.3	20.7	27.1	20.1	26.6	19.6	26.2	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	16.2	22.4	18.2	15.4	22.1	17.9	15.0	21.9	66.1	26.5	64.0	26.8	62.2	26.2	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.9	-1.6	36.8	1.4	1.7	-2.7	38.0	-3.5	39.0	-4.3	39.9	-5.4	41.2		
(5)			-3.7	21.9	1.3	1.2	-4.6	22.8	-5.4	23.5	-6.1	24.1	-7.1	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	23.9	23.5	21.9	18.7	15.6	12.9	12.5	15.3	19.2	21.9	22.9	23.9
	DBStd	4.97	2.45	2.16	2.29	2.65	2.68	2.50	2.55	3.30	3.39	3.29	3.12	2.65
	HDD10.0	20	0	0	0	0	2	6	8	3	0	0	0	0
	HDD18.3	608	1	0	2	26	89	163	182	102	30	9	4	1
	CDD10.0	3422	430	378	368	261	176	93	85	168	277	368	388	431
	CDD18.3	968	172	144	113	37	4	0	0	8	57	119	141	173
	CDH23.3	10161	1570	1268	986	406	150	31	24	207	799	1425	1559	1735
(13) Wind	WSAvg	3.6	3.7	3.4	3.1	3.0	2.9	3.1	3.3	3.9	4.2	4.5	4.4	4.1
	PrecAvg	562	118	84	75	52	16	5	2	6	14	44	69	88
	PrecMax	829	317	242	187	187	78	44	12	57	69	134	210	235
	PrecMin	350	28	15	4	0	0	0	0	0	0	1	8	2
	PrecStd	118	62	49	45	43	21	9	4	11	18	33	50	51
	DB	36.2	35.0	34.0	31.0	28.8	26.1	25.7	29.2	33.1	35.2	35.8	36.7	
	MCWB	17.2	17.0	16.5	15.8	13.1	11.4	10.6	12.8	13.9	14.9	15.5	17.0	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	33.9	33.1	31.9	29.0	26.2	23.9	23.9	27.2	31.5	33.8	33.9	34.2	
	2% MCWB	17.6	17.4	16.9	15.4	12.8	10.7	10.0	11.9	13.6	14.7	15.6	17.2	
	5% DB	32.1	31.4	30.1	27.2	25.0	22.3	22.2	25.8	30.0	32.1	32.2	32.3	
	5% MCWB	17.8	17.6	17.2	15.6	12.4	10.1	9.8	11.5	13.2	14.7	15.7	17.2	
	10% DB	30.2	29.9	28.5	25.9	23.7	21.0	20.8	24.0	28.1	30.3	30.8	30.9	
	10% MCWB	18.0	18.0	17.2	15.3	12.1	9.8	9.4	10.9	12.8	14.3	15.9	17.3	
	WB	21.5	21.5	20.7	19.1	16.1	13.5	12.5	14.7	16.7	18.7	19.9	20.7	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	27.8	26.8	26.6	25.6	23.3	21.0	21.2	24.8	26.9	27.6	27.6	28.1	
	2% WB	20.6	20.6	19.8	18.1	14.8	12.2	11.5	13.6	15.5	17.5	19.1	20.1	
	2% MCDB	27.2	26.5	25.8	24.6	22.2	20.3	20.1	23.8	26.5	27.2	27.2	27.8	
	5% WB	20.0	20.0	19.2	17.3	13.9	11.5	10.7	12.6	14.6	16.7	18.3	19.5	
	5% MCDB	26.3	26.0	25.5	23.8	21.4	19.7	19.8	22.6	25.9	26.6	26.8	27.2	
	10% WB	19.5	19.4	18.6	16.5	13.1	10.6	10.0	11.7	13.7	16.0	17.6	19.0	
	10% MCDB	25.8	25.7	24.9	22.8	20.8	18.9	19.5	21.6	25.1	26.0	26.2	26.7	
(35) Mean Daily Temperature Range	MDBR	11.8	11.8	12.2	12.9	15.0	15.8	15.8	16.2	15.9	14.6	13.7	12.6	
	5% DB	14.7	14.6	15.0	15.0	17.0	17.6	17.5	17.5	17.6	16.7	15.8	15.0	
	5% MCWBR	4.7	4.9	5.2	5.7	7.1	8.0	8.2	7.8	7.5	6.6	5.5	5.1	
	5% WB	11.1	11.1	11.5	12.1	13.6	14.7	15.1	15.3	14.9	13.5	12.8	12.3	
	5% MCWBR	4.2	4.2	4.5	5.3	6.3	7.3	7.8	7.6	6.9	5.7	5.0	4.8	
(40) Clear-Sky Solar Irradiance	taub	0.365	0.372	0.356	0.323	0.285	0.269	0.268	0.322	0.369	0.394	0.346	0.353	
	taud	2.398	2.376	2.406	2.456	2.534	2.549	2.526	2.365	2.238	2.213	2.403	2.423	
	Ebn at Noon	977	955	941	928	927	924	938	914	912	923	990	992	
	Edh at Noon	128	128	118	102	85	80	85	110	137	149	126	125	
(44) All-Sky Solar Radiation	RadAvg	7.06	6.49	5.84	4.93	4.46	4.01	4.37	5.18	6.27	6.83	7.25	7.32	
	RadStd	0.71	0.58	0.47	0.41	0.18	0.17	0.10	0.23	0.24	0.37	0.56	0.45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page